Rocks and Minerals
ANSWER KEY

Brown Science Olympiad Division B
Invitational December 7, 2025



Station 1

1) Identify the following rock. Bauxite (+1)



2) What type of rock is it? Be as specific as possible naming potential subtypes (not just igneous, sedimentary, or metaphoric) Chemical sedimentary (+2, +1 for correct rock type)

3) What two metal elements are these rocks usually associated with? Aluminum and Iron (+2)

4) Where in the United States is the rock mostly mined for? Arkansas (+1)

5) Explain the concept of laterization and how it relates to this rock. **Tiebreaker**

Intense rainfall and heat cause leaching, where water dissolves and carries away soluble materials. What's left behind are insoluble oxides of aluminum and iron. If the original rock contains aluminum-bearing minerals, then over long periods of laterization, the aluminum accumulates while other components wash away. The end result is a bauxite deposit. (+3)

6) Outline a possible pathway aligning with the rock cycle that might lead to the formation of this rock. Start from the initial formation of igneous rocks from the solidification of molten rock and go from there. Words or diagrams are fine. (+2) for including formation of igneous rocks, specifically naming granite,gneiss, basalt, syenite, or shale, and how the rock is chemically weathered and leached to form bauxite.

Station 2

A

1) Identify the mineral. Chalcopyrite (+1)



2) What two other minerals are flanking the mineral in 1)? Dolomite and Quartz (+2)
3) What class of mineral is it? Sulfide (+1)
4) What is the chemical formula of this mineral? CuFeS2 (+1)
5) What is the approximate hardness of this mineral? 3.5, 3-4 is fine too (+1)
6) What would you expect to see if you performed a hardness test on the mineral using a knife? What about a copper coin? Knife will scratch Chalcopyrite, copper coin and chalcopyrite shouldn't scratch each other since both have the same hardness (+2)
7) What element is this mineral a common ore of? Copper (+1)

B

1) Identify the mineral Celestite (+1)

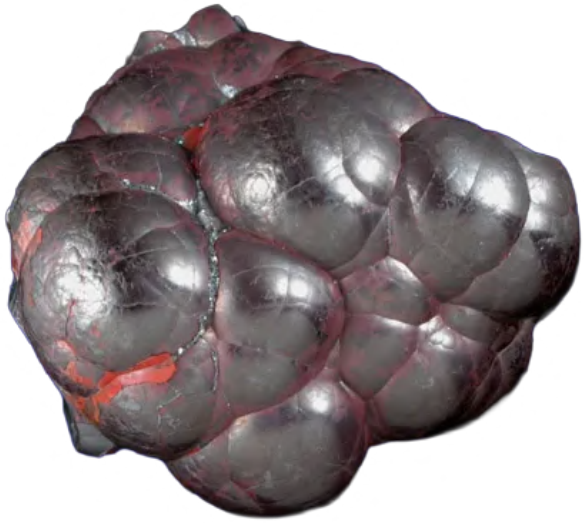


2) What family does this mineral belong to? Sulfates (+1)
3) This mineral is often used in a particular item because of a certain element it contains that can produce a specific color. Name the item that uses this mineral and the element. Fireworks (+1), Strontium (+1)
4) Frequently this mineral is blue in color. What property of the mineral gives off the blue color? Lattice Defects (+1)

Station 3

1) Identify the following mineral (+1) Hematite

2) What distinct crystal habit does the image display? Botroyidal (+1)



3) What is the streak of this mineral? Accept red of any kind (+1)
4) Is this mineral magnetic? (Y/N) No (+1)
5) What is the hardness of this mineral? 5-6 (+1)
6) Where in the United States is this mineral most commonly found? Minnesota, Lake Superior (+1)
7) What common sedimentary rock/feature commonly displays this mineral? Banded Iron Formations (+1)
8) How does this mineral pertain to the exploration of Mars? What does the presence of this mineral on Mars indicate about the planet?
   Hematite is often used to select landing spots for Martian probes. (+1) Typically on Earth, Hematite is a good indication of water being present, which is why hematite is particularly interesting in the exploration of Mars as a possible indication for life and evidence that water existed on Mars (+2)

Station 4

A

1) Identify the following. Lepidolite (+1)



2) What type of fracture and cleavage does this mineral exhibit? Uneven fracture, micaceous cleavage (+2)
3) What family does this mineral belong to? Mica (+1)
4) What two other minerals on the rocks & minerals list are the most similar to this mineral? Biotite and Muscovite (+2)
5) What element is this mineral particularly rich in? Lithium (+1)
6) Which igneous rock is this mineral often found in? Granite (+1)
7) Based on your previous answer or your knowledge of the mineral, is this mineral likely felsic or mafic in origin? Felsic (+1)

B

1) Identify the following mineral. Sphalerite (+1)



2) What is the chemical formula of this mineral? (Zn,Fe)S (+1)
3) What family does this mineral belong to? Sulfides (+1)
4) What is the luster of this sample of the mineral? What other luster does this mineral commonly display? Adamantine (+1), Resinous (+1)
5) Based on your answer for the previous question, give a chemical explanation for why these two lusters exist. Determine what other element gives the mineral its particular appearance. The presence of iron determines whether sphalerite is dark/adamantine or translucent/resinous in luster (+1). Thus the mineral provided is rich in iron (+1).
6) What element is this mineral a common ore of? Zinc (+1)
7) What do you expect to see if you shine the mineral under UV light? The mineral should fluoresce and produce varying colors. Color itself depends on the impurities in the mineral. (+1)

Station 5

1) Identify the mineral Rutile (+1)



2) What is the chemical formula of this mineral? TiO2 (+1)
3) What type of rock are you most likely to find this mineral? What about the environment is favorable to the formation of this mineral? Metamorphic (+1), High temp/pressure environment (+1)
4) This mineral displays a unique feature of being able to pierce other minerals and rocks. What do you call this structure when it forms in quartz? Rutilated Quartz (+1)
5) What is the name of the crystal habit in the previous question? Acicular (+1)
6) This mineral frequently displays twinning. What kind of symmetry allows for the twinning in this mineral? Twining depends greatly on the geometry of the crystals. Rutile forms tetragonal lattices which allow for the geometry of twinning (+1), must mention tetragonal
7) What are the 6 different symmetries/systems that can occur in different minerals that can allow twinning? **Tiebreaker** Triclinic, Monoclinic, Orthrombic, Tetragonal, Hexagonal, Isometric (+0.5 each)
8) Describe the difference between contact and penetration twinning. Contact twinning occurs when two crystal individuals share a single, planar twin boundary and meet along that flat surface. Penetration twinning involves two intergrown crystals that intersect in a more complex, often interpenetrating geometry with a non-planar twin boundary. (+2)

Station 6

A)

1) Identify the following Amazonite (+1), if answer Microcline/orthoclase only give (+0.5)



2) What is the chemical composition of the mineral? KAlSi3O8
3) Explain the concept of polymorphism and connect it to this mineral. **Tiebreaker**
   Polymorphism in minerals is when a single chemical compound can exist in more than one crystal structure. These different forms, called polymorphs, have the same chemical composition but different physical properties due to their varied arrangements of atoms, which are determined by the temperature and pressure under which they formed (+2). Microcline/orthoclase is a high-temperature polymorph of Amazonite
4) Does this mineral exhibit perfect cleavage (Y/N)? Y (+1)
5) What element gives this mineral its green/blue color? What can you use to confirm the presence of this element? Lead (+1), UV radiation will cause fluorescence (+1)

B)

1) Identify the following Albite (+1)



2) What broad classification fits both this mineral and the previous mineral? Feldspars (+1)
3) This mineral primarily has more of which element compared to that of the previous mineral? More Sodium (+1), Microcline has more potassium but otherwise chemical composition is the same.
4) What environments are favorable for the formation of this mineral? Based on your answer, what types of rocks will typically show this mineral? **Tiebreaker** Albite will typically form in low temp and low pressure environments (+2 for both low pressure and temp). Felsic, igneous (+1)
5) Both of these minerals exhibit what kind of symmetry that allows for twinning? Triclinic system (+1)

Station 7

1) Identify the following Olivine (+1)



2) This mineral forms in what types of rocks? Mafic or Ultramafic (+1)
3) What is the streak and hardness of this mineral? White (+1), 7 (+1)
4) What temperature does this mineral crystallize at in celsius? 1250-1350 Celsius, anywhere in that range is fine (+1)
5) Fill in the blank. This mineral reacts with __ to create __. Silica (+1), Pyroxene (+1).
6) Describe the concept of fractional crystallization. Which elements are removed early on and which elements remain after a long period of time in this process? Fractional crystallization is the process in which early-formed minerals crystallize from a cooling magma and are removed from the melt as they settle by gravity, causing the remaining liquid to change composition over time. (+1) Early-forming minerals remove Mg, Fe, and Ca, while the melt becomes progressively enriched in Si, Na, and K as crystallization continues. (+1)
7) What's the difference between the continuous and discontinuous reaction series? Be as specific as possible and describe when the continuous and discontinuous reaction series converge. **Tiebreaker** Plagioclase Feldspar begins crystallizing roughly at the same time as pyroxene on the Discontinuous Series. At this temperature, Plagioclase Feldspar is calcium-rich, but it becomes more sodium-rich as temperature decreases. The series is continuous because the mineral is always Plagioclase Feldspar. (+2, +1 for naming the first mineral and change in composition as temperature decreases) The discontinuous series is the process described in question 4) where olivine goes to pyroxene then amphibole to biotite then potassium feldspar as temperature decreases (+1 for describing process). At potassium feldspar, the discontinuous and continuous pathways converge (+1).
8) Name two subtypes of this mineral and describe briefly how they differ chemically. Fayalite, iron rich (+1) Forsterite, magnesium rich

Station 8

1) What two minerals are present in the sample? Malachite and Azurite (+2)



2) What is the chemical composition of both minerals? Cu2CO3(OH)2 for malachite (+1) Cu3(CO3)2(OH)2 for azurite (+1)
3) Will this mineral react with acid at low temperatures? What about high temperatures? Malachite will react with acid at low and high temperatures (+1)
4) How are these two minerals related when they occur together and how can one mineral lead to the formation of the other? The two represent different stages of the same geochemical process: azurite forms under slightly higher CO2 conditions, while malachite forms when CO2 is lower (+1) Azurite is unstable in air and is usually pseudo-morphically replaced by malachite (+1)
5) These minerals are good indicators of what deposit? Copper deposits (+1)
6) What's the hardness of both minerals? Both around 3.5-4 (+1)
7) What crystal habit does the specimen show? Botroiydal (+1)

Station 9

Minerals

1) Identify the following Calcite (+1)



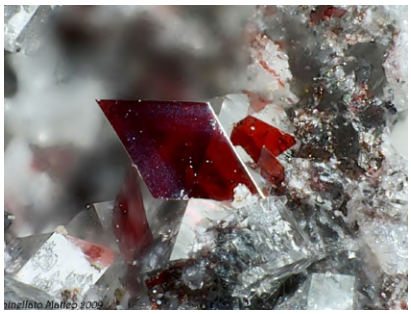
2) Identify the following Barite (+1)



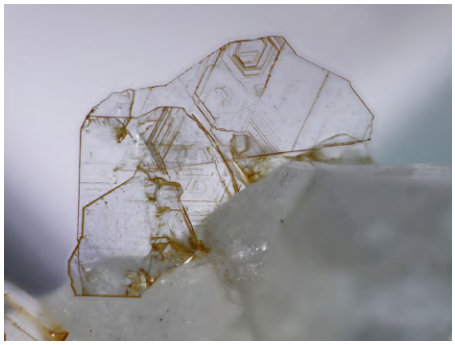
3) Identify the following Dolomite (+1)



4) Identify the following Hematite (+1)



5) Identify the following Talc (+1)



6) Identify the following Corundum (+1)



7) Identify the following Aragonite (+1)



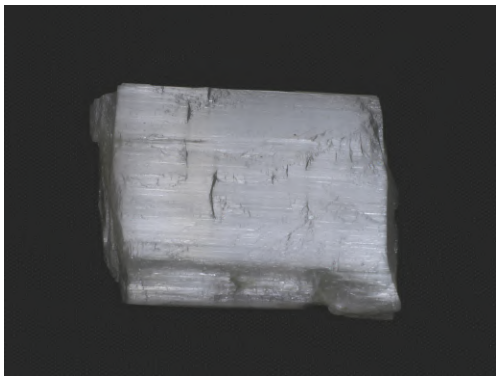
8) Identify the following Copper (+1)



Station 10

Minerals

1) Identify the following Ulexite (+1)



2) Identify the following Beryl (+1)



3) Identify the following Apatite (+1)



4) Identify the following Willemite (+1)



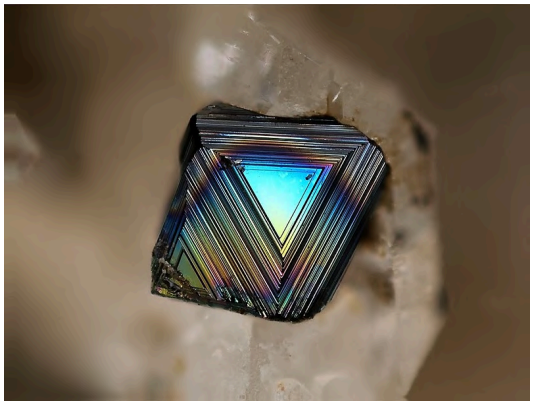
5) Identify the following Tiger's Eye (+1)



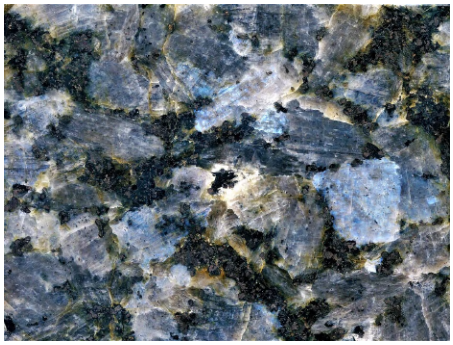
6) Identify the following Staurolite (+1)



7) Identify the following Magnetite (+1)



8) Identify the following Labradorite (+1)



Station 11

Minerals

1) Identify the following Chrysocolla (+1)



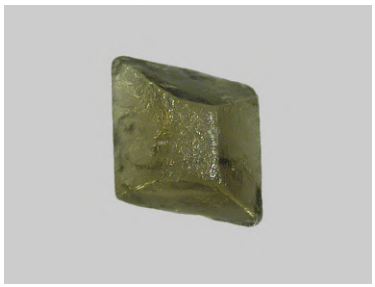
2) Identify the following Turquoise (+1)



3) Identify the following Galena (+1)



4) Identify the following Augite (+1)



5) Identify the following Magnesite (+1)



6) Identify the following Sodalite (+1)



7) Identify the following Fluoraparite (+1)



8) Identify the following Kyanite (+1)



Station 12

Rocks

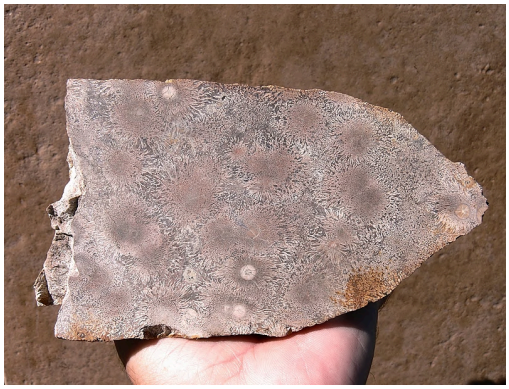
1) Identify the following Obsidian (+1)



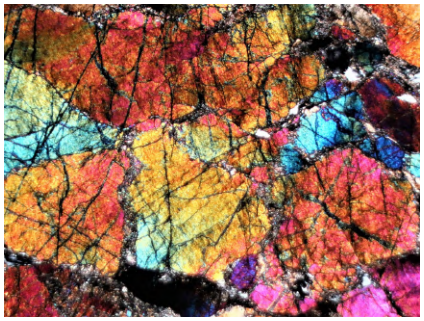
2) Identify the following Pegmatite (+1)



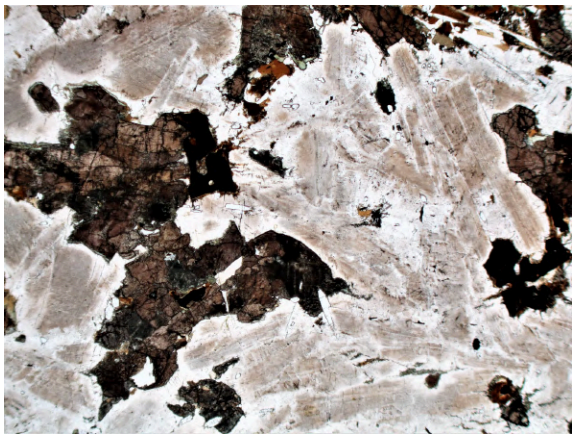
3) Identify the following Rhyolite (+1)



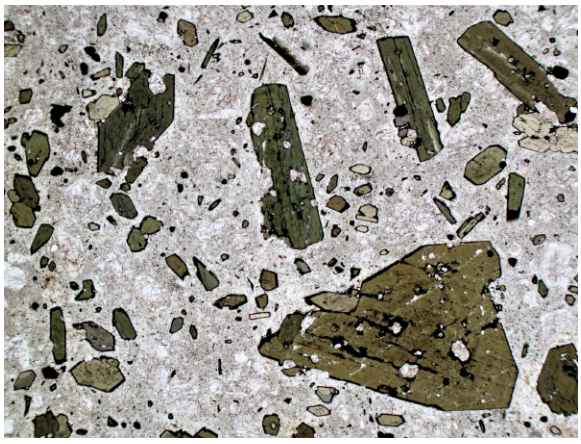
4) Identify the following Peridotite (+1)



5) Identify the following Gabbro (+1)



6) Identify the following Andesite (+1)



7) Identify the following Tuff (+1)



8) Identify the following Granite (+1)



Station 13

Rocks

1) Identify the following Dolostone (+1)



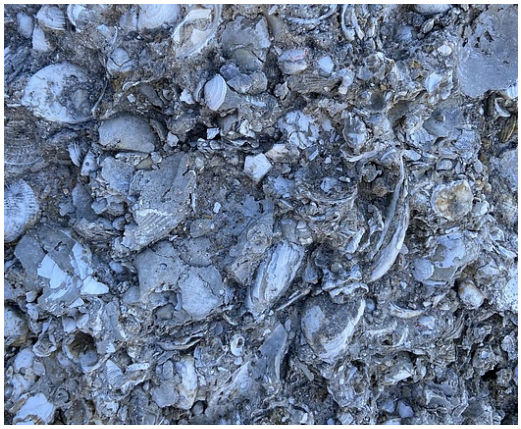
2) Identify the following Shale (+1)



3) Identify the following Lignite (+1)



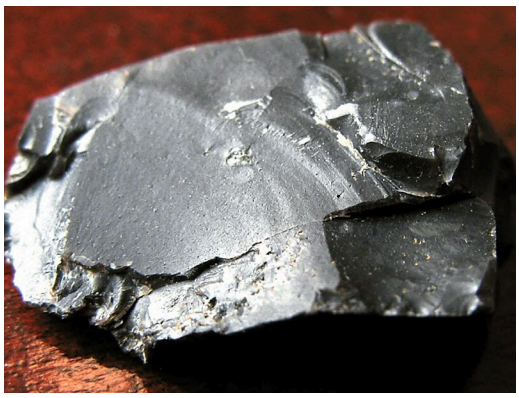
4) Identify the following Coquina (+1)



5) Identify the following Diatomite (+1)



6) Identify the following Chert (+1)



Station 14

Rocks

1) Identify the following Phyllite (+1)



2) Identify the following Gneiss (+1)



3) Identify the following Quartzite (+1)



4) Identify the following Mica Schist (+1), only 0.5+ for schist



5) Identify the following Slate (+1)

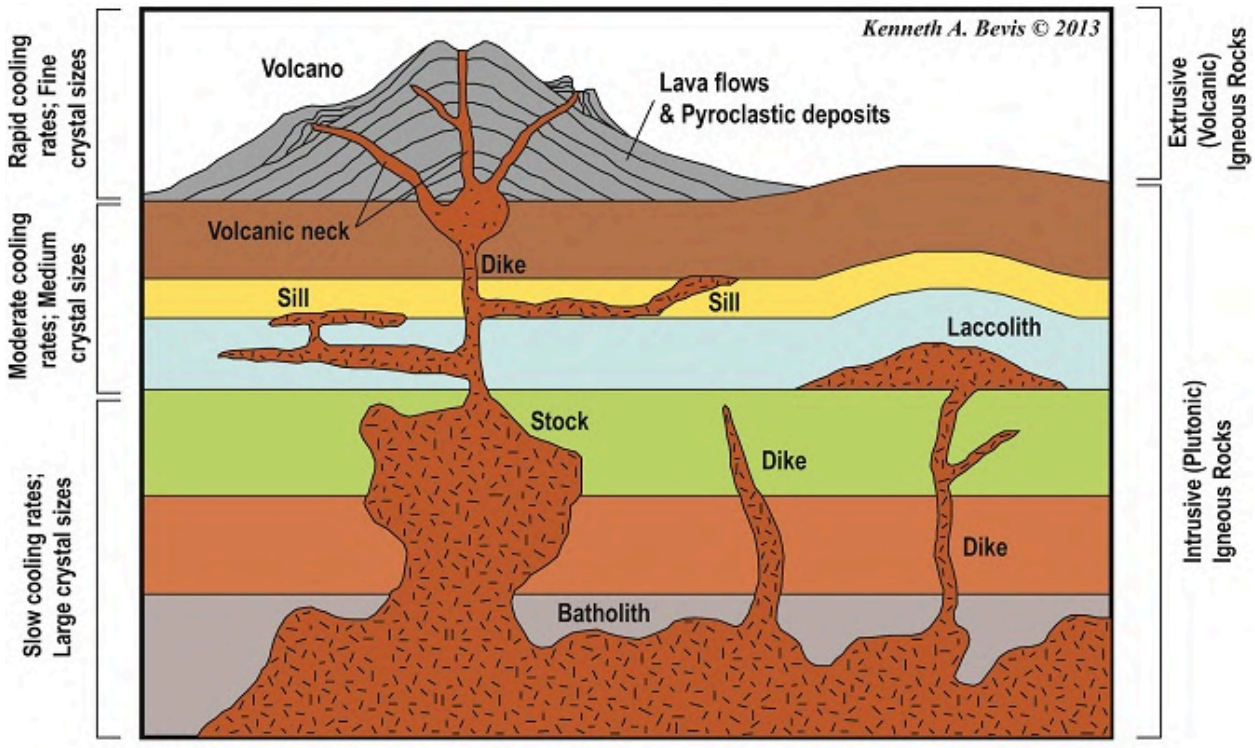


6) Identify the following Serpentine (+1)



Station 15

1) Label the following diagram 0.5+ for each correct, 3.5 points total

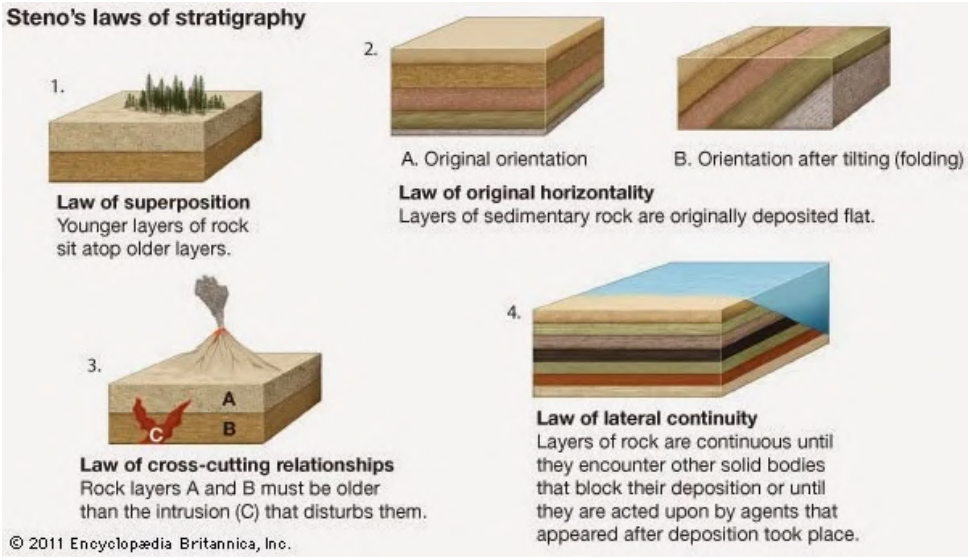


2) Define what a Xenolith is. A xenolith is a fragment of rock that becomes trapped inside a different rock, most commonly when older rock is torn off and incorporated into magma, which then hardens into an igneous rock, +1 point

3) You discover a tabular igneous body that is concordant with sedimentary bedding but has baked contacts both above and below. What type of intrusion is it? **Tiebreaker** +1 Sill

Station 16

1) Name and briefly explain the four major principles used in relative dating For each of the four, 0.25+ for correct name and 0.25+ for correct description. +2 points total



2) Fill in the following diagram

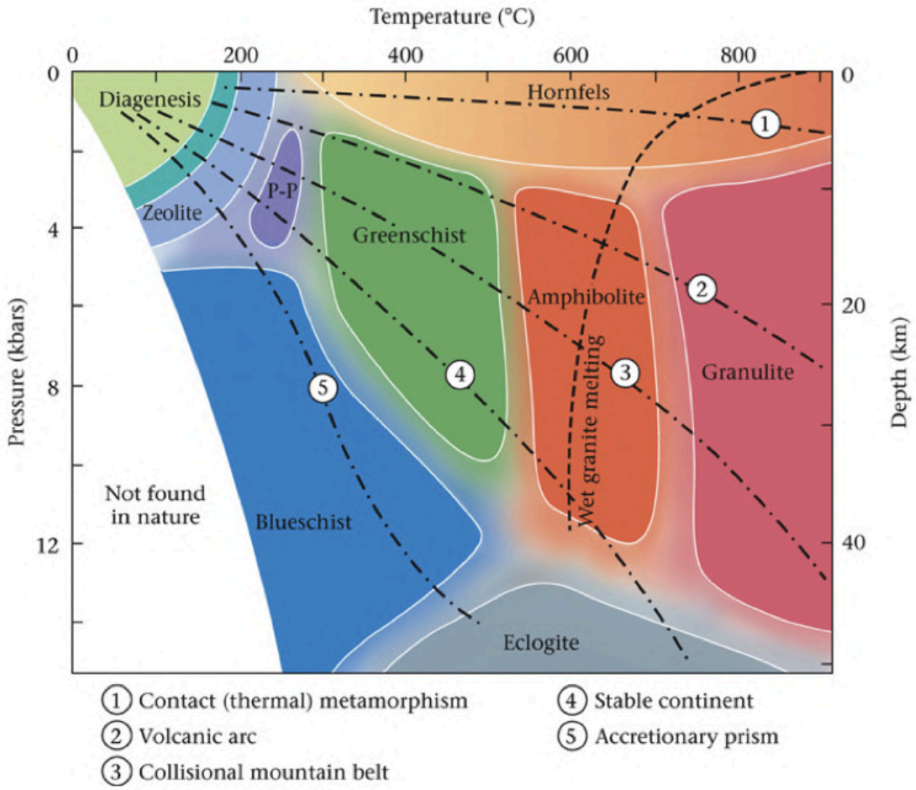
Answer (Oldest to Youngest): K J M E F L D C B A I H G

If pair of two are swapped but everything else is correct, -2
If pair of four are swapped but everything else is correct, -4
If pair of six or more are swapped/incorrect, -6
If completely incorrect/blank, -8

8 points possible

Station 17

1) Label the following diagram 1+ point for each



2) Hornfels < greenschist < blueschist < amphibolite < granulite, 2.5 points total

-1 if two are ordered wrong

-2 if four are ordered wrong

-2.5 if totally incorrect/blank

Station 18

Label the following 0.5+ point for each correct, +8 points total

Terrestrial Environments

Alluvial Fan (#7) — a deposit shaped like an open fan formed at the base of mountains where streams suddenly disgorge from a narrow valley and spread out, dumping their sediment load.

Dune (#5) — a wind-deposited accumulation of sand-sized material found in deserts and along beaches.

River Channel (#9) — the long, narrow depression in which a stream or river is confined during normal periods of flow.

Flood Plain (#8) — an area adjacent to a river which is covered by water during times of flooding and underlain by river-deposited sediment.

Lake (#16) — a body of relatively still water localized in a basin, surrounded by land.

Glacier (#6) — a persistent body of dense ice that is constantly moving under its own weight, abrading rock and debris from their substrate to create various landforms as they move.

Transitional Environments

Delta (#10) — a stream deposit near the mouth of a river where it flows into a larger body of water. Deltas may include features within them such as swamps and tidal flats.
A **tidal flat** is a low area adjacent to the sea that is affected by tides—submerged during high tide, exposed during low tide; consists of both channels and non-channel areas.

Lagoon (#11) — an oceanic to freshwater environment protected by the obstruction of an offshore bar or barrier.

Bar (#12) — an environment of shifting sand-sized sediment deposited offshore from a beach; or an elongated accumulation of sand (or gravel) in streams and rivers.

Beach (#4) — the transition zone between the land and a large body of water, e.g. lake or ocean, typically where waves break onto the shore; zone of sand deposition.

Marine Environments

Reef (#3) — a rock, sandbar, or biological construction lying beneath the surface of the water.

Continental Slope (#15) — the steep change of slope from the continental shelf to the abyssal plain; usually consists of mud and silts.

Continental Shelf (#13) — the part of the continent that extends offshore beneath the sea; the relatively shallow water body overlying it is known as a shelf sea.

Abyssal Plain (#2) — an underwater plain on the deep ocean floor, usually at depths between 3000 and 6000 m.

Submarine Canyon (#14) — a steep-sided valley cut into the sea floor of the continental slope and shelf.

Submarine Fan (#1) — marine equivalent of an alluvial fan; sediments derived from the continental shelf and slope.

Brown Invitational - Repaired practice edition - Answer key

Division B | Rocks And Minerals | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

s2-10. (2 points) Authored replacement for source item s2-10: Supplied field notes place celestite with gypsum in evaporite beds. Describe an environment consistent with those observations.

answer: An evaporative basin where dissolved sulfate minerals precipitate as water becomes concentrated; an arid restricted marine basin is one suitable example. Use the supplied field notes, not the unrelated fireworks answer. (2 points).

s2-11. (1 points) Authored replacement for source item s2-11: Supplied crystallographic data: chalcopyrite is tetragonal and celestite is orthorhombic. Do these two specimens belong to the same crystal system?

answer: No. Tetragonal and orthorhombic are different crystal systems. (1 points).

s6-3. (2 points) Authored replacement for source item s6-3: Repair data: green amazonite is a variety of microcline; microcline is triclinic and orthoclase is monoclinic; both feldspars have composition KAlSi3O8. Explain which pair illustrates polymorphism.

answer: Microcline and orthoclase have the same composition and different crystal structures, so they illustrate polymorphism. Amazonite is a variety of microcline, not a separate polymorph contrasted with microcline. (2 points).

s6-5. (2 points) Authored replacement for source item s6-5: Can a green feldspar’s color or ultraviolet appearance alone establish that it contains lead? Describe a stronger test.

answer: No. Color/fluorescence is not a uniquely diagnostic elemental measurement. Use a suitable calibrated chemical/elemental analysis and standards rather than inferring lead solely from an appearance. (2 points).

s15-1. (4 points) Authored replacement for source item s15-1: Label these supplied descriptions: A is a sheet parallel to bedding; B cuts across bedding; C is a concordant dome-shaped intrusion; D is an intrusive body with exposed area greater than 100 square kilometres.

answer: A: sill. B: dike/dyke. C: laccolith. D: batholith. Award one point for each correct label. (4 points).

s16-1. (2 points) Authored replacement for source item s16-1: Name and explain these four relative-dating principles: superposition, original horizontality, lateral continuity and cross-cutting relationships.

answer: Superposition: younger beds overlie older beds in an undeformed sequence. Original horizontality: sediments are initially deposited approximately horizontally. Lateral continuity: beds originally extend laterally until thinning or reaching a boundary. Cross-cutting: the cutting feature is younger than what it cuts. Award credit for the meaning of each specified principle. (2 points).

s17-1. (3 points) Authored replacement for source item s17-1: Replace the missing facies diagram using this supplied schematic: A has moderate pressure and lower temperature; B has similar pressure and higher temperature; C has high pressure and low temperature. The supplied candidate facies are greenschist, amphibolite and blueschist. Match each once.

answer: A: greenschist. B: amphibolite. C: blueschist. These are qualitative positions in a supplied schematic, not exact pressure-temperature boundaries. (3 points).

s17-2. (3 points) Authored replacement for source item s17-2: Explain why hornfels, greenschist, blueschist, amphibolite and granulite cannot all be put into one universal lowest-to-highest temperature grade sequence without pressure information.

answer: Facies depend on pressure as well as temperature. Blueschist is characteristically high-pressure/low-temperature, while hornfels is associated with low-pressure contact metamorphism. At comparable moderate pressure, greenschist-amphibolite-granulite can express increasing temperature, but it is not a universal ordering of all listed facies. (3 points).